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Git Hooks

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What are Git Hooks?

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Git hooks are scripts that run automatically when certain Git events happen, like making a commit or pushing code.

Why Use Hooks?

Hooks help you automate repetitive tasks, enforce coding standards, and catch problems early.

For example, you can:

- Run tests before every commit or push
- Check code style automatically
- Block bad commit messages
- Enforce rules for everyone on your team

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By default, you'll see sample scripts ending with `.sample`.

Example: List Available Hooks

```
[user@localhost] $ ls .git/hooks
```

How to Enable a Hook

To enable a hook, remove the `.sample` extension and make the script executable.

For example, to enable `pre-commit`:

Example: Enable pre-commit Hook (Linux/macOS)

```
[user@localhost] $ mv .git/hooks/pre-commit.sample .git/hooks/pre-commit
[user@localhost] $ chmod +x .git/hooks/pre-commit
```

On Windows, just rename the file to `pre-commit` and make sure it can be run by your shell (e.g. use `.bat` or `.ps1` if needed).

REMOVE ADS

Types of Hooks

There are many types of hooks, but the most common are:

- `pre-commit`

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`post-receive`

pre-commit Hook

The `pre-commit` hook runs before you make a commit.

You can use it to check code style, run tests, or stop a commit if something is wrong.

Example: Simple pre-commit Hook

```
#!/bin/sh
# Stop commit if any .js file has "console.log"
grep -r 'console.log' *.js && {
  echo "Remove console.log before committing!"
  exit 1
}
```

commit-msg Hook

The `commit-msg` hook checks or edits the commit message.

For example, it can block commits without a ticket number.

Example: commit-msg Hook

```
#!/bin/sh
# Block commit if message does not contain a ticket number
if ! grep -qE 'JIRA-[0-9]+' "$1"; then
  echo "Commit message must have a ticket number (e.g. JIRA-123)"
```

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pre-push Hook

The `pre-push` hook runs before you push code to a remote.

You can use it to run tests or checks before sharing code.

Example: pre-push Hook

```
#!/bin/sh
npm test || exit 1
```

Server-side Hooks

Some hooks (like `pre-receive`) run on the Git server, not your computer.

These can enforce rules for everyone who pushes to the repository.

Example: pre-receive Hook

```
#!/bin/sh
# Block pushes to main branch
grep refs/heads/main || exit 1
```

Custom Hooks

You can write any custom script as a hook.

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```
#!/bin/sh
echo "Hello from my custom hook!"
```

Debugging and Best Practices

- Make sure your hook script is executable (`chmod +x scriptname`).
- Add `echo` statements to see what your script is doing.
- Check the exit code: `exit 0` means success, `exit 1` means fail.
- On Windows, use `.bat` or `.ps1` scripts if needed.
- Keep hooks simple and fast-slow hooks slow down your workflow.
- Share useful hooks with your team (but remember: hooks are not versioned by default).

Note: Hooks are powerful for automating checks (like linting or tests) and enforcing team standards. Client-side hooks run on your computer. Server-side hooks run on the Git server.

Exercise ?

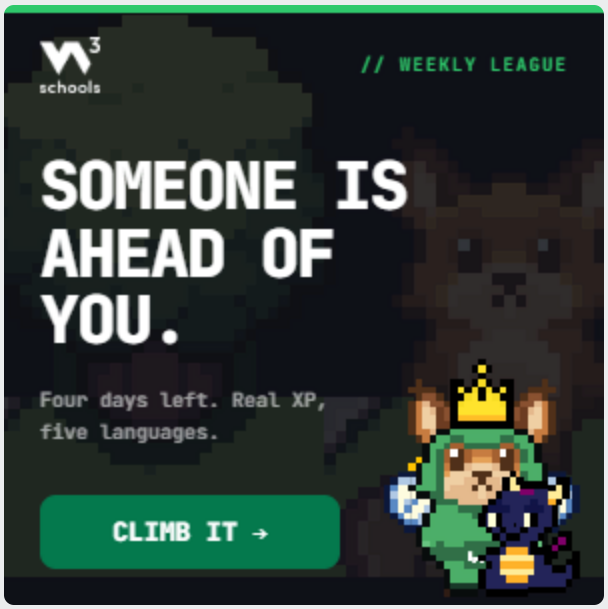
Drag and drop the correct command to search for "console.log" in all .js files in a pre-commit hook.

```
#!/bin/sh [          ] -r 'console.log' *.js
```

ls grep chmod

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